

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - II, April 2023

MA1004-1 – DISCRETE MATHEMATICS & NUMERICAL METHODS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Using Taylor's series method up to 4 th power of x , compute $y(0.2)$. Given $y' = 1 - 2xy$, $y(0) = 1$ and $h = 0.2$.	5	L*2	3	2
	b) Find a real root of $2x - \log_{10}x - 7 = 0$ in $(3, 4)$, using Regula falsi method. Carry out 3 iterations.	5	L3	3	1
2.	a) Prove that Newton-Raphson method has second order convergence.	4	L2	3	1
	b) Given that $\frac{dy}{dx} = 3x + y$, $y(0) = 1$ and $h = 0.1$. Find an approximate value of $y(0.2)$ by Runge-Kutta method of order 4.	6	L3	3	2
Unit – II					
3.	a) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:				
		5	L3	4	2
	b) Find the solution of the equation $u_{xx} = 2u_t$ with $u(0,t) = u(4,t) = 0$ and $u(x,0) = x(4-x)$, taking $h = 1, k = 1$. Find the values up to $t = 3$.	5	L1	4	2
4.	a) Solve the PDE $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0 = y$; $x = 3 = y$ with $u = 0$ on the boundary and $h = 1$.	5	L3	4	2
	b) Evaluate the pivotal values of the equation $u_{tt} = u_{xx}$, taking $h = k = 0.2$ up to $t = 0.4$. The boundary conditions are $u(0,t) = u(1,t) = 0$, $u_t(x,0) = 0$ & $u(x,0) = \frac{1}{2}x(1-x)$.	5	L1	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
